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9

MU and Quantification Limits

The ability of a method to quantify working samples at low analyte concentration is an important topic of discussion because once a limit is established, it should facilitate method selection. Very often, it is also used as a commercial argument. Most commonly, the quantification capacity threshold is defined through two parameters, the limit of detection (LOD) and the limit of quantification (LOQ), even though many other parameters are also proposed in the analytical literature.

LOD is defined by metrologists as “the measured value, obtained by a given measurement procedure, for which the probability of falsely claiming the absence of a component in a material is, given a probability of falsely claiming its presence” (International Vocabulary of Metrology [VIM]). It applies to any measuring method and is not limited to analytical sciences.

This definition is compatible with statistical significance testing that involves two risks of error, usually named α and β (in this context, β must not be confused with the β probability of the β -expectation tolerance interval β -ETI). These error risks are classically fixed in the statistical literature at 1% or 5%, respectively.

It should be recalled that *detecting* an analyte leads to a qualitative result expressed by presence/absence, conforming/nonconforming, or yes/no. LOD theoretically represents the starting point where the instrumental signal is perceptible but not translated into a quantity of matter.

On the other hand, LOQ is specific to analytical sciences and quantitative methods. Unfortunately, there is no single standardized definition, either in VIM or in International Union of Pure and Applied Chemistry (IUPAC). It would rather correspond to an observed parameter derived from various extrapolations or interpolations. For quantitative methods, the real issue is the LOQ. It is therefore, this performance limit that is discussed here.

For qualitative methods, one of the parameters introduced to evaluate a method's performance limit is the equivalency of the level of detection (LD) accompanied by a probability (i.e. for example, LD50). Unfortunately, this abbreviation LD can lead to confusion with the LOD.

9.1 Definitions and Assessment of LOQ

For the LOQ, a definition and a calculation method were already proposed in Section 5.2.2, many other definitions exist and, consequently, many estimation methods. An attempt at a rough general definition can be proposed, and several variations are available in the guides, as explained in a recent review [1]. The most classical definition is the following, and in square brackets, suggested details are added:

“The lowest concentration of an analyte that can be quantified [in a specific matrix] with acceptable precision and accuracy [and under specified operating conditions].”

But also:

“The concentration for which the risks of error α and β are acceptable.”

This creates confusion with the LOD when the risk levels are not clearly specified. Ultimately:

“The value for which the bias and relative standard deviation [or coefficient of variation] of repeatability do not exceed a given acceptance threshold.”

From a methodological point of view, various calculation methods have been proposed. A succinct description may help to understand the practical problems raised by each procedure. Some decades ago, a well-documented historical review on the LOQ parameter was published but, unfortunately, not recently updated [2].

9.1.1 Multiple Blank Standard Deviations

In the 1970s, blank standard deviation was the method recommended by the Environmental Protection Agency (EPA). The simplest solution is to calculate the LOD and LOQ as k -fold standard deviation of the intercept a_0 of the calibration curve (assumed being a straight line) denoted s_{a_0} and given by Eq. (2.18). This parameter is assumed to be sufficiently comparable to the standard deviation s_{bl} of a blank sample.

Whatever the k -fold value, the result is expressed in the same units as the instrumental signal. For instance, the peak area must be converted into a concentration by dividing by the slope a_1 (which corresponds to the sensitivity).

A classic methodological error is the use of a calibration curve that would neglect the matrix effects, the influence of the selected calibration model, or the possible heterogeneity of variances of the measurements at the different concentrations (or heteroscedasticity). Inverse calibration is a well-known operation to convert an instrumental response into a concentration.

Because the coefficient a_1 is an estimation of the slope, it may vary from one series to another, and computed LOD or LOQ may also vary. Moreover, if the intercept a_0 is significantly different from zero (or even negative), another source of variability

is present, which is not always easy to consider. For the LOD, a multiplicative coefficient k_D is applied to the standard deviation s_{a_0} . It varies according to the validation guide, for example $k_D = 3.0, 3.3, 6.0$, etc. These values approach Normal distribution law quantiles used in statistical significance tests but do not strictly conform to the theory.

$$\text{LOD} = k_D \times \frac{s_{a_0}}{a_1}$$

For LOQ, a similar principle remains, and by convention, the EPA proposed to take a multiplicative coefficient $k_Q = 10$, without any statistical justification, just according to a general principle that implies that the LOQ must be “sufficiently distant” from the LOD. Finally, this approach often leads to unrealistic values but is nevertheless accepted by control authorities.

$$\text{LOQ} = k_Q \times \frac{s_{a_0}}{a_1}$$

9.1.2 Visual Examination

Accepted in the pharmaceutical industry [3] and in biomedical analysis, this method relies on analyst's expertise who knows the discrimination capability of the method he applies to expect a correct quantification. The justification of the threshold is provided by signal records, such as a chromatogram. In this case, the LOQ is directly expressed in the unit of the analyte, i.e. in absolute or relative concentration. The drawback is the subjective aspect of the procedure. If an obtained value is used for commercial promotion, it may be uneasily arguable.

9.1.3 Signal-to-Noise Ratio

This approach is based on the physical theory of instrumental signals. It mainly applies to spectroscopic methods, such as atomic emission, gamma-ray spectrometry, nuclear magnetic resonance, ultraviolet (UV)–Visible. The procedure often advocated is to compare the signals obtained with low-concentration samples to those from blanks and determine the minimum concentration for which the analyte can be detected or quantified.

A signal-to-noise ratio $S/N = 3 : 1$ is generally considered acceptable for a LOD and 10:1 for a LOQ. The same values as previously assigned to k_D and k_Q . It should be noted that in the context of separative methods, quantification is often carried out based on integrated peak area. In this case, a LOQ obtained from the signal-to-noise ratio does not make sense because the word “ratio” does not refer to the area but to the signal height.

Therefore, this approach has serious drawbacks with modern instrumentation, where the signal is digitized. The baseline signal is usually filtered and smoothed, and it is difficult, if not impossible, to measure a true instrumental background. Hence, the evolution of recent instruments, such as high-resolution mass spectrometers, has resulted in extremely low background noise, and this approach has become pointless. Some discussion about the inconsistency of the signal-to-noise ratio with this kind of method is available in [4].